RB: Resonant Bifurcation — Overcoming Cognitive Interference via Dynamic Topological Fission

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RB: Resonant Bifurcation — Overcoming Cognitive Interference via Dynamic Topological Fission

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> **STATUS: SUPERSEDED by RB+G** (`rbg.md`), which adds generational capping, an alias ledger, energy-conserving fan-out, explicit readout and an audit trail, and reports 20-seed reproducibility data. This paper is retained as the **precursor record**: it states the original mechanism and the reasoning that produced it, without the controls that later proved necessary. Cite `rbg.md` for the mechanism as implemented; cite this only for priority of the idea.

Abstract

When a concept accumulates contradictory structure, a wave-propagation substrate representing it as a single cavity suffers destructive interference: opposed waves cancel inside the cavity and the plasticity updates they would have produced cancel with them. Conventional systems respond by re-weighting connections. We propose instead that the engine **physically splits the node**.

On a sustained high-incoherence, high-tension signal the substrate executes a structural mutation: it clones the cavity, routes excitatory tunnels to one child and inhibitory tunnels to the other, and binds the children with a single high-stiffness tunnel so the system continues to treat them as one concept while allowing them to oscillate independently.

The thesis is that intelligence does not only re-weight existing connections — it **spawns new structure to house concepts its current structure cannot hold**.

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1. Trigger Condition

The incoherence ratio I is conventionally logged as a failure metric. Here it is used as a **trigger**.

Fission fires when, over three consecutive structural ticks:

$$I > 0.15 \quad \text{and} \quad \tau > \tau_{\text{threshold}}$$

The interpretation is that the concept is **too complex to be represented by a single point**. It contains contradictions, and its representation is the wrong shape rather than the wrong value.

Requiring persistence over three ticks distinguishes a structural condition from a transient: a single noisy tick should not restructure the graph.

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2. The Mitosis Protocol

When the trigger fires, the engine performs a structural mutation in three steps.

Clone. Create `X_alpha` and `X_beta` from parent concept `X`.

Polarity routing. Route every excitatory tunnel (positive Σ^+) to `X_alpha`, and every inhibitory tunnel (negative Σ^-) to `X_beta`.

Concept binding. Connect `X_alpha` and `X_beta` with a single ultra-high-stiffness tunnel — a *semantic spring*.

The binding is what prevents fission from being mere deletion of a concept. The children remain mechanically coupled, so the substrate still treats them as the same thing, while their displacements are no longer forced to share one variable.

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3. Why It Works

Before fission, positive and negative waves arrive at one cavity and interfere destructively. Both the observable displacement and, more damagingly, the Hebbian velocity product are driven toward zero — so the tunnels carrying the contradictory evidence are never strengthened.

After fission they enter **two separate resonant cavities** that are bound together. Taking a worked case:

- `metformin_alpha` accumulates the positive associations — *treats diabetes*, *activates AMPK*.
- `metformin_beta` accumulates the negative ones — *causes GI distress*, *inhibits complex I*.

Because the semantic spring is stiff, the engine still knows they are the same entity. Because they are distinct cavities, they oscillate independently and no longer cancel each other's plasticity.

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4. Why This Is Difficult on Standard Infrastructure

The mechanism requires **dynamic graph topology**, and mainstream deep learning infrastructure is built to forbid it.

Matrix multiplication requires fixed dimensions. Splitting a node changes the adjacency matrix from $N \times N$ to $(N{+}1) \times (N{+}1)$ *during* a run. Frameworks organised around static tensor shapes

and pre-compiled kernels treat that as an error condition, not a capability.

A localised wave-propagation model written against ordinary memory has no such constraint: a new cavity is an allocation and a few pointer updates. **The mechanism is unavailable not because it is a bad idea but because the dominant substrate cannot express it** — which is a claim about tooling, not about cognition.

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5. What RB+G Added, and Why It Was Necessary

The mechanism above is incomplete in ways that only appear on contact with a real ontology. `rbg.md` supplies:

Addition	Problem in the original
Generational cap	children can themselves fission, without a bound the graph grows without limit
Alias ledger	after splitting, a query for `metformin` must still resolve; the original specifies no mapping from parent to children
Energy-conserving fan-out $(1/\sqrt{2})$	cloning tunnels duplicates energy, injecting it into the field from nothing
Explicit readout	with children present, what a query *returns* is unspecified above
Audit trail	a structural mutation with no provenance record cannot be explained or undone
Auto-tuned trigger	$I > 0.15$ is hand-picked for one ontology; the successor derives it from field energy
20-seed measurement	this paper reports no measurement at all

The last row is the substantive one. The mechanism as stated here is a *proposal*; RB+G is the version that was measured against a fission-disabled baseline across 20 seeds and four ablation levers.

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6. Conclusion

Resonant Bifurcation's contribution is the reframing: a persistently incoherent concept indicates that the **representation has the wrong shape**, and the correct response is to change the structure rather than the weights. That claim survived into RB+G unchanged.

Everything else in this paper — the specific threshold, the routing rule, the absence of energy accounting — was superseded within one iteration. The idea was right and the controls were missing, which is the ordinary condition of a first draft and the reason this one is labelled as a precursor rather than presented as a result.

References

- Mahir, I. *RB+G: Gated Recursive Bifurcation with Analogical Bridging.* `rbg.md` — **the citable version of this mechanism.**

- Mahir, I. *Synaptic Receptor Gating (SRG).* `srg.md` — the static-topology alternative to fission.
- Mahir, I. *Recursive Integrity Field Architecture (RIFA).* — source of the incoherence and tension signals used as the trigger.
- Implementation: `src/layer2.hpp` (`fission_locked`).